



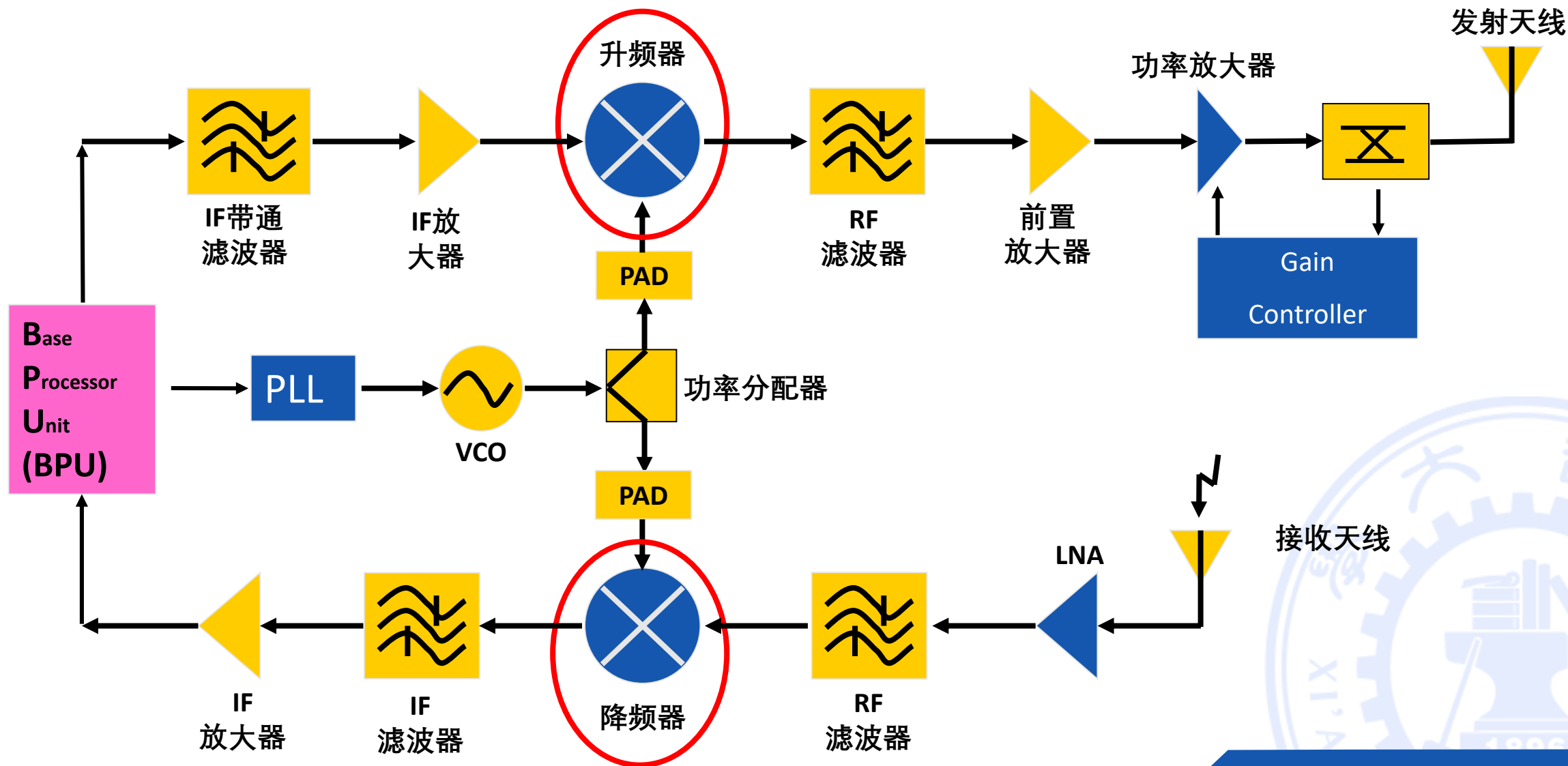
西安交通大学
XI'AN JIAOTONG UNIVERSITY

实验八 通信系统链路仿真



一、 基本理论

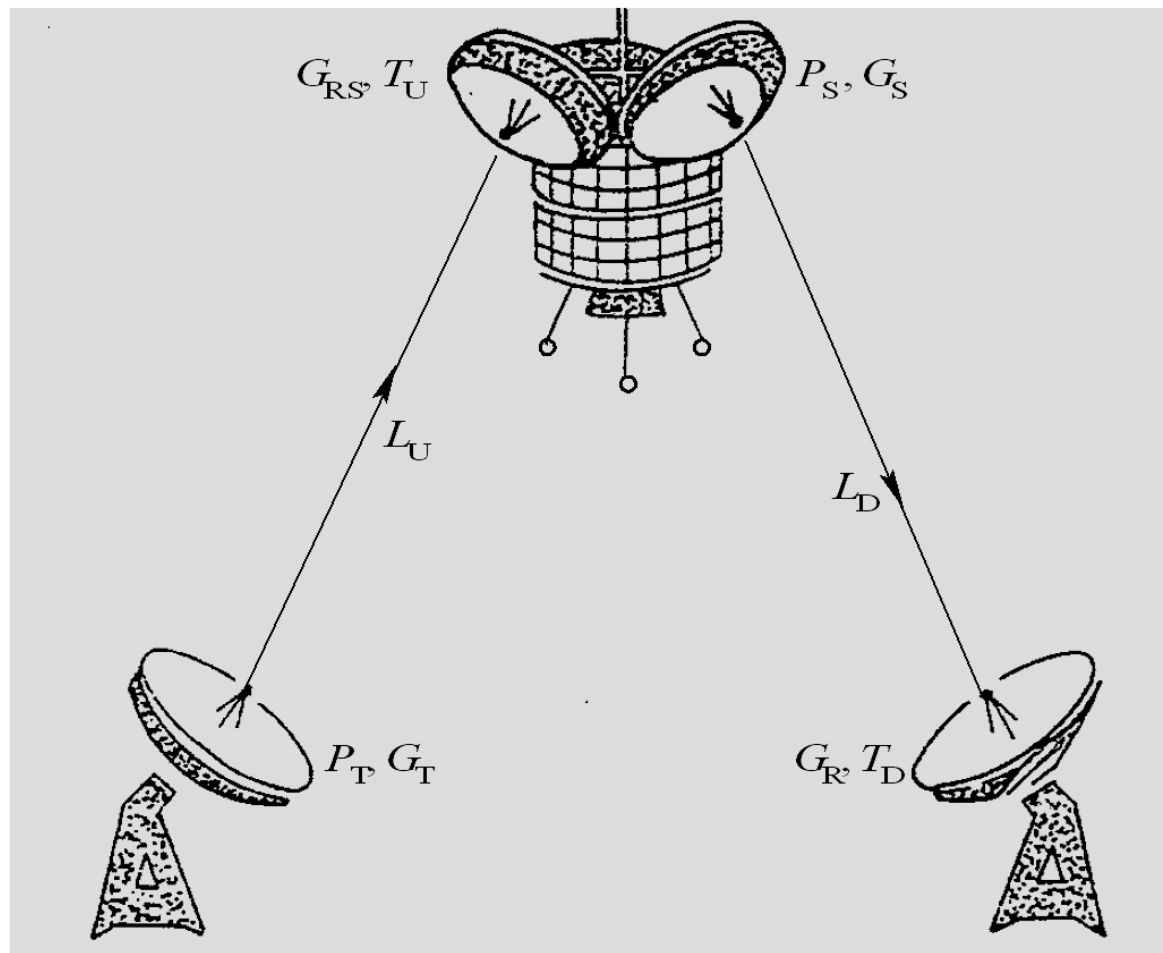
1. 无线通信链路的组成：发射系统、信道和接收系统





2. 无线通信链路设计计算目的:

有效地在两个通信点之间提供可靠而又高质量的联接手段

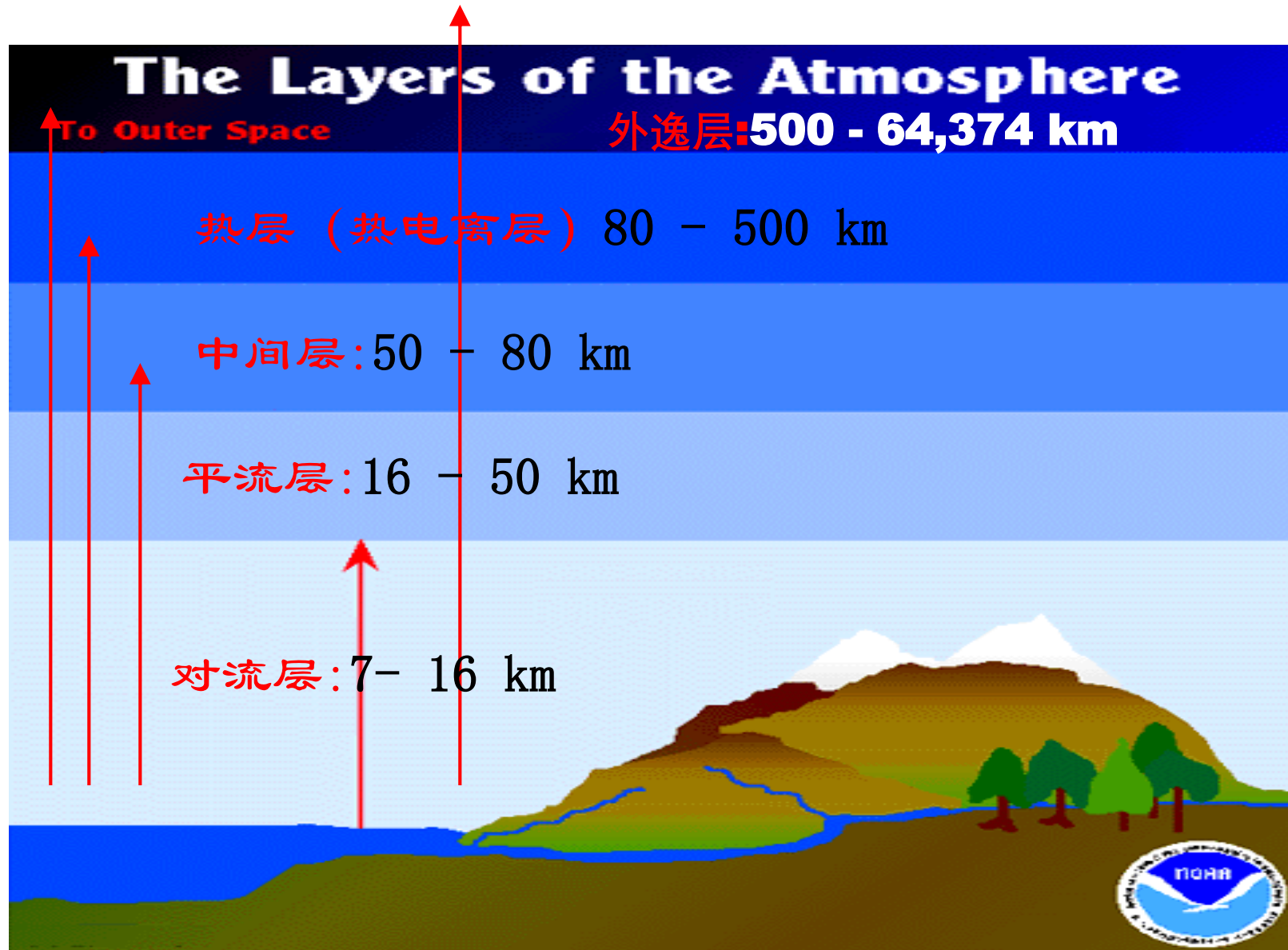


单向空间链路一般示意图

T_U 为上行链路噪声、 T_D 为下行链路噪声温度； L_D 下行线路传输损耗、 L_u 上行线路传输损耗； G_R 接收天线增益、 G_T 发射天线增益； P_T 发射天线功率。



卫星通信的电波要经过对流层（含云层和雨层）、平流层、电离层和外层空间，跨越距离大，影响电波传播的因素很多。





3. 指标

对于模拟信号传输是以解调后的信噪比S/N来表示；对数字信号传输则用比特误码率(BER)表示。

以卫星通信为例，链路预算主要考虑两方面的问题：

- 已知通信卫星和地球站的电参数， 计算通信链路的传输能力。
- 已知卫星的电参数， 根据对传输容量和质量的要求， 确定地球站的设备参数

➤ 噪声温度 $N = kT_e B$

➤ 噪声系数

➤ 灵敏度：

➤ 1dB增益压缩点

➤ 三阶交调损耗

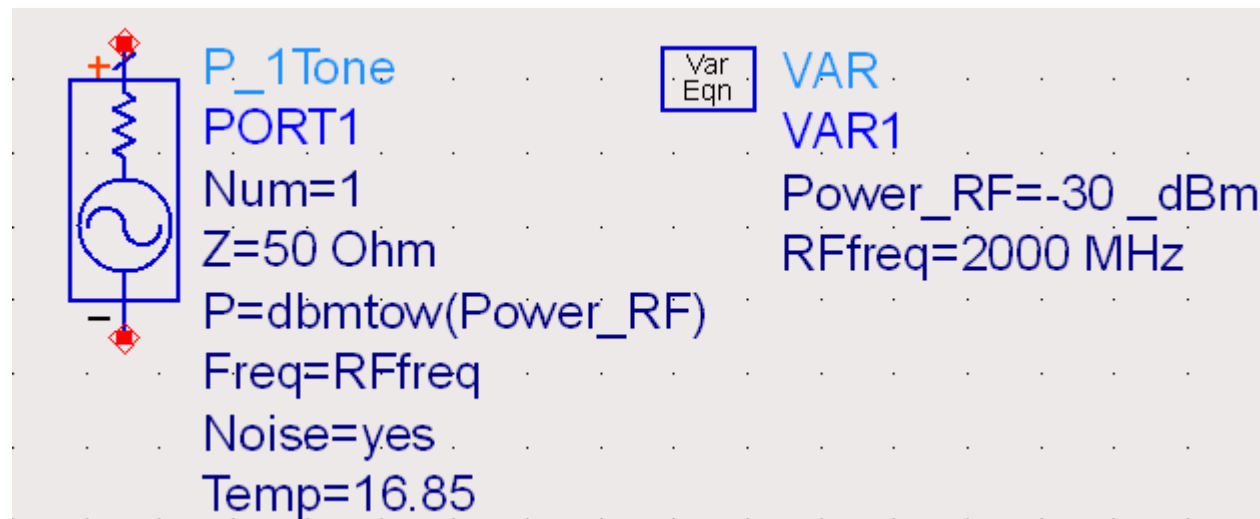
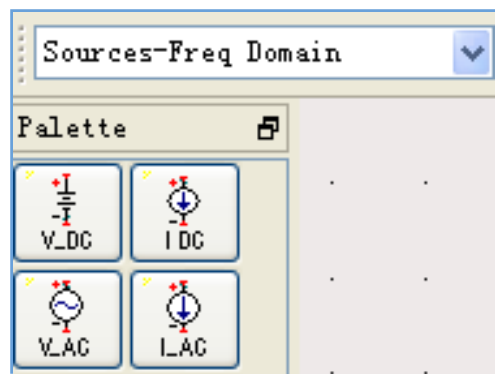




二、简单系统的链路预算示例

1. 新建一个工程

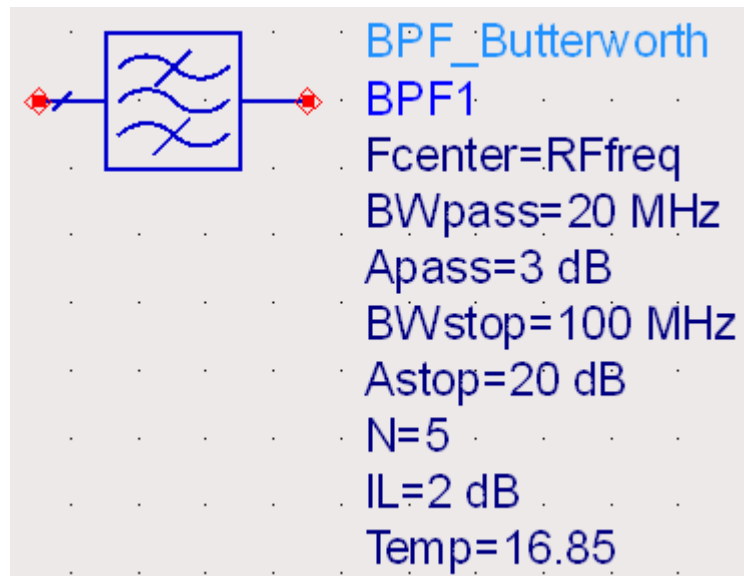
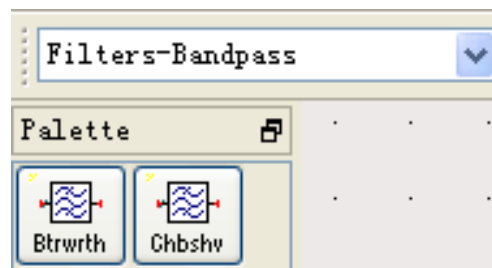
- 选择“P_1Tone”控件添加到原理图中
- 单击VAR控件，添加变量“Power_RF”和“RFfreq”，并设置其值如下





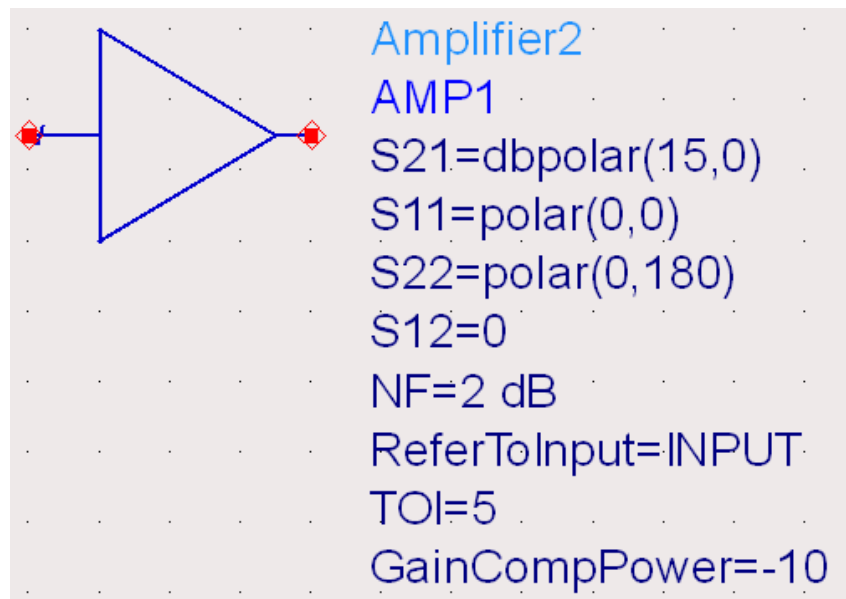
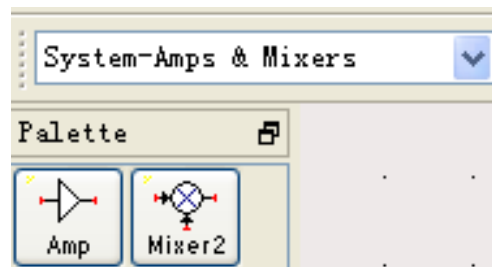
2. 第一级滤波器

- 添加BPF_Butterworth滤波器并设置



3. 第一级放大器

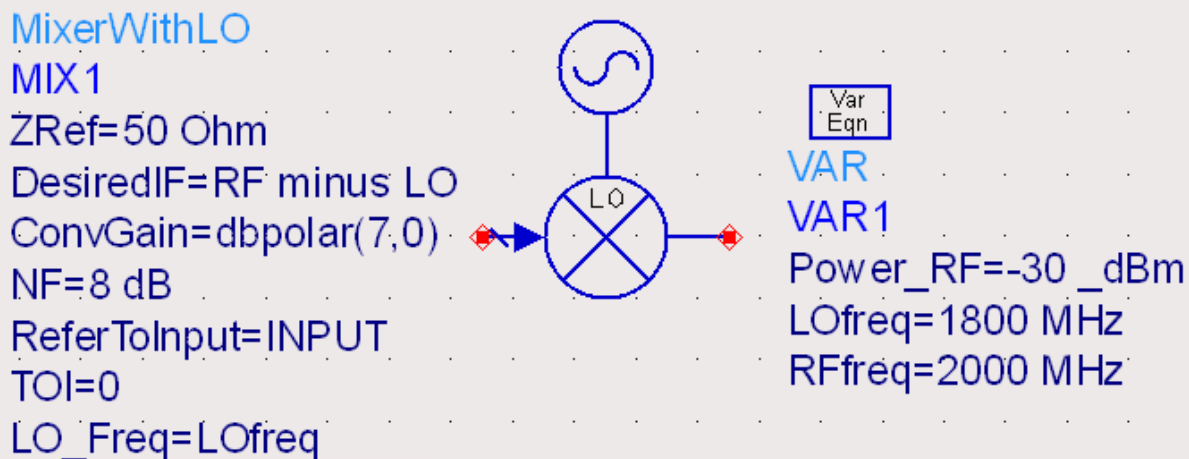
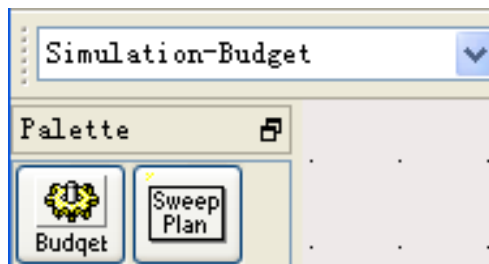
- 添加放大器控件





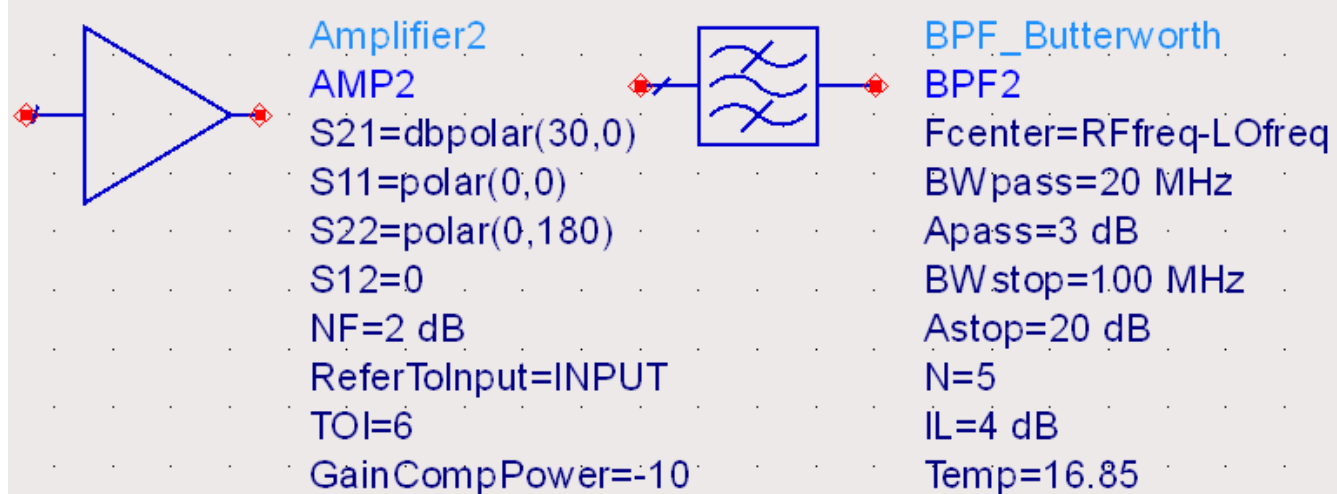
4. 本振及混频

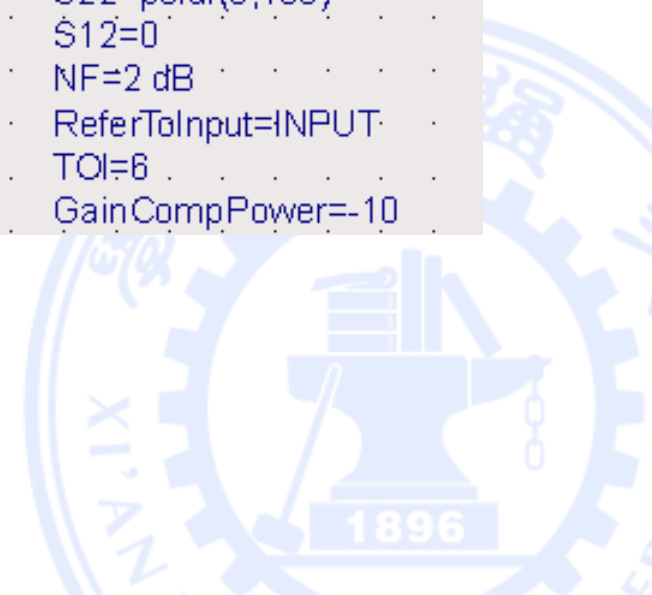
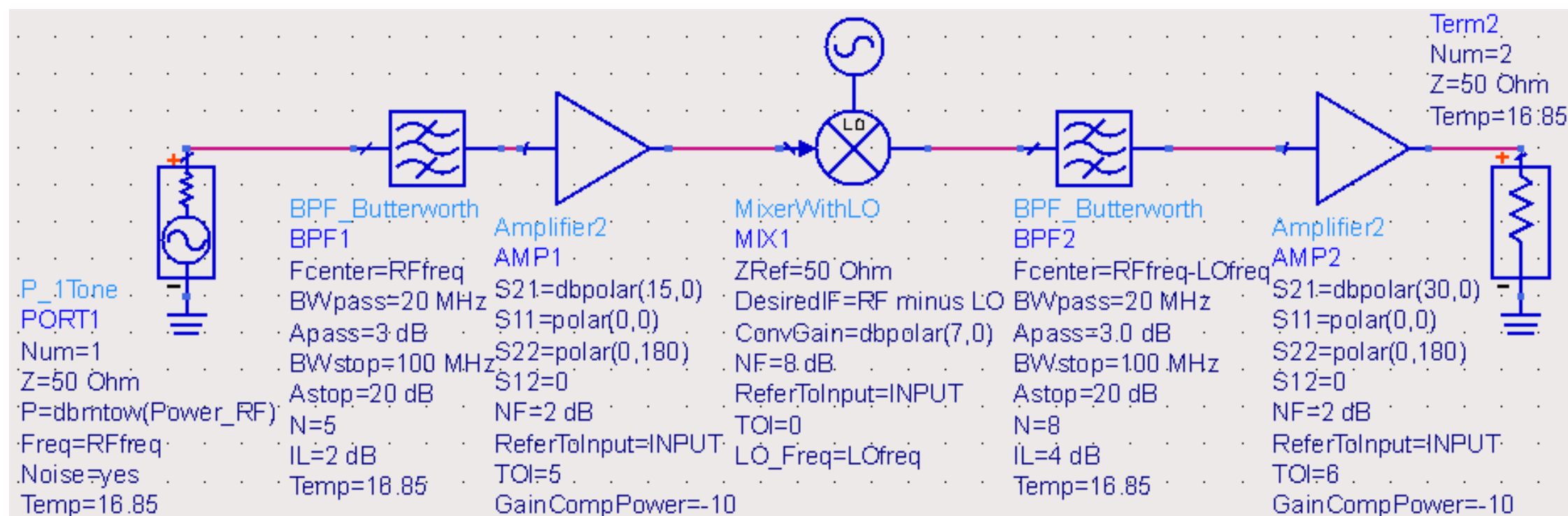
- 添加MixWLO控件, 设置参数
- 在VAR控件中, 添加本振频率变量LOfreq



5. 添加第二级滤波器、第二级放大器

- 复制第一级滤波器、第一级放大器, 设置参数, 添加端口2, 完成系统

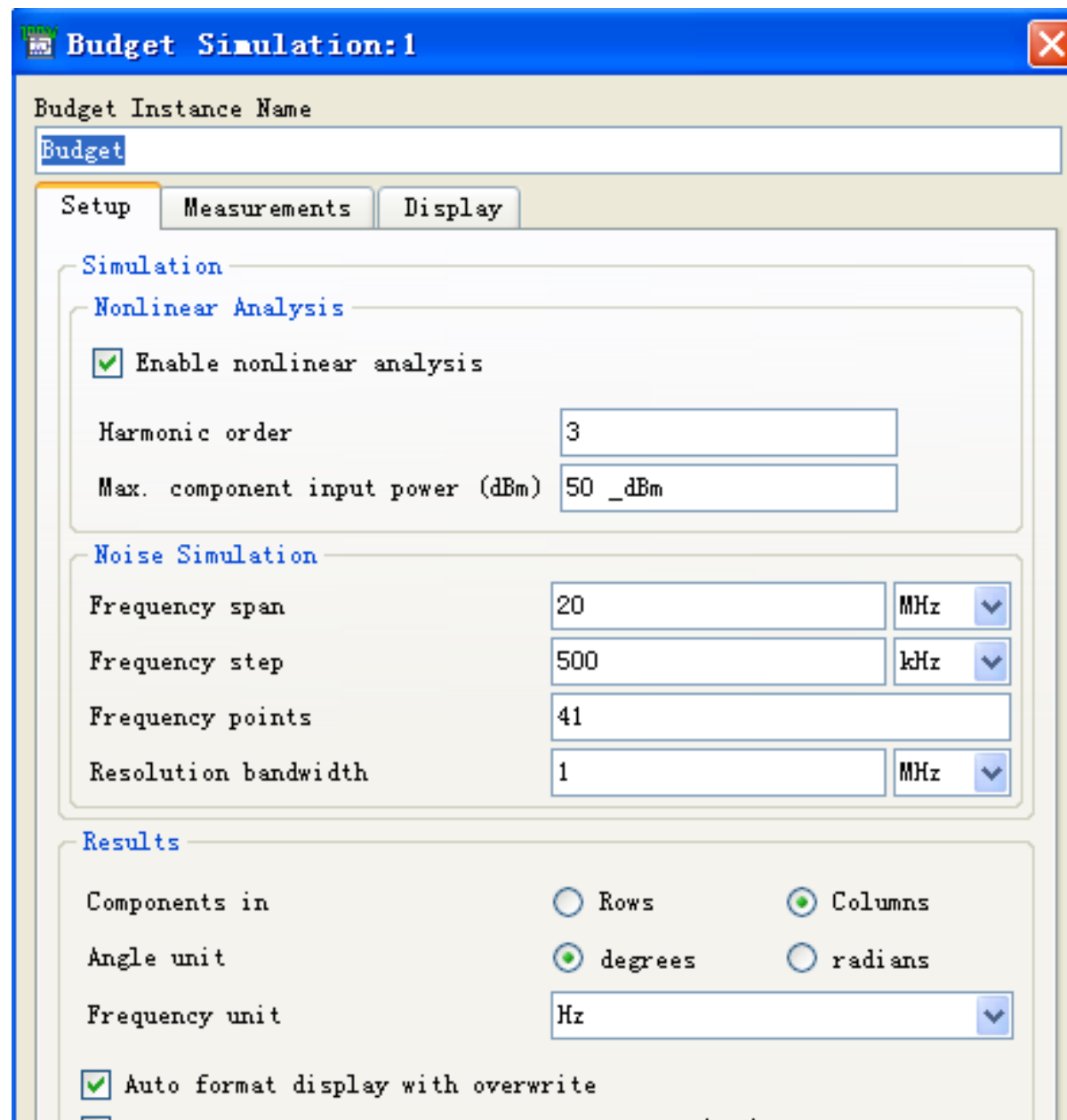
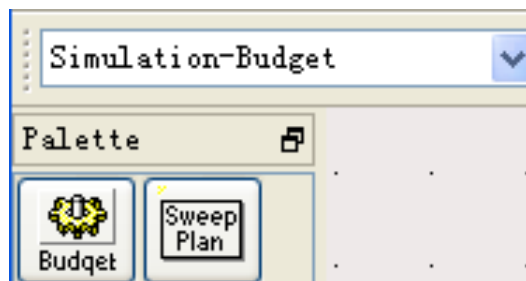






6. BUDGET控制器设置

➤ 添加Budget控件





Setup Measurements Display

Recall selected measurements from file

Available Measurements

- Cmp_Ctrb_SysNF_NoI
- Cmp_Ctrb_SysTOI_dB
- Cmp_LS_GainChange_
- Cmp_OutNO_dBm
- Cmp_OutP1dB_dBm
- Cmp_OutSOI_dBm
- Cmp_S11_dB
- Cmp_S11_mag
- Cmp_S11_phase
- Cmp_S12_dB
- Cmp_S12_mag
- Cmp_S12_phase

>>Add>>

<<Delete<<

Selected Measurements

- Cmp_NF_dB
- Cmp_S21_dB
- Cmp_OutTOI_dBm
- NF_RefIn_NoImage_dB
- OutNPwrTotal_dBm
- OutP1dB_dBm
- OutPGain_dB
- OutPwr_dBm
- OutSFDR_Total_dB
- OutSNR_Total_dB

Raise Lower

Description





BUDGET

Budget

Budget

NonlinearAnalysis=yes

NonlinearHarmonicOrder=3

CmpMaxPin=50 _dBm

NoiseFreqSpan=20 MHz

NoiseFreqStep=500 kHz

NoiseResolutionBW=1 Hz

TableComponentFormat=Columns

MeasurementFrequencyUnit=Hz

MeasurementAngleUnit=degrees

AutoFormatDisplay=yes

OutputCSVFile=no

RunCommand=no

SystemCommand=

Measurement[1]="Cmp_NF_dB"

Measurement[2]="Cmp_S21_dB"

Measurement[3]="Cmp_OutTOI_dBm"

Measurement[4]="NF_Refln_NoImage_dB"

Measurement[5]="OutNPwrTotal_dBm"

Measurement[6]="OutP1dB_dBm"

Measurement[7]="OutPGain_dB"

Measurement[8]="OutPwr_dBm"

Measurement[9]="OutSFDR_Total_dB"

Measurement[10]="OutSNR_Total_dB"

Measurement[11]="OutTOI_dBm"





7. 仿真结果及分析

Meas_Name	BPF1	AMP1	MIX1
Cmp_NF_dB	2.000	2.000	8.000
Cmp_S21_dB	-2.000	15.000	7.000
Cmp_OutTOI_dBm	1000.000	20.000	7.000
NF_Refln_NoImage_dB	2.000	4.000	4.437
OutNPwrTotal_dBm	-101.215	-84.100	-75.230
OutP1dB_dBm	1000.000	4.000	-3.688
OutPGain_dB	-2.000	12.998	19.823
OutPwr_dBm	-32.000	-17.002	-10.177
OutSFDR_Total_dB	1000.000	69.400	54.791
OutSNR_Total_dB	69.215	67.098	65.053
OutTOI_dBm	1000.000	20.000	6.957

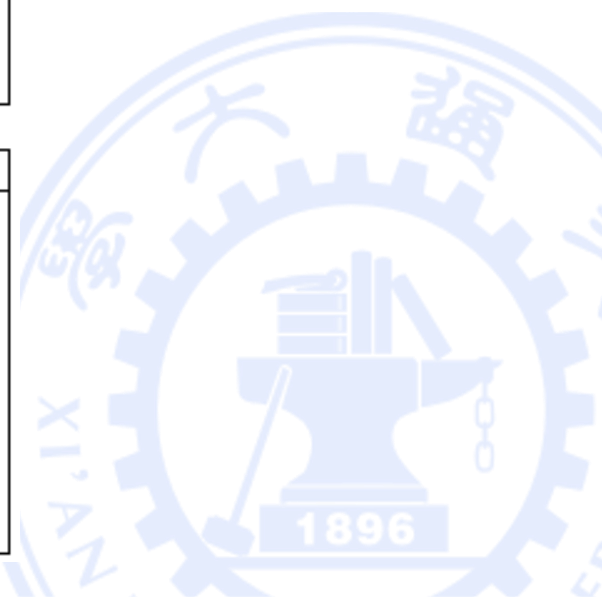
BPF2	AMP2
4.000	2.000
-4.000	30.000
1000.000	36.000
4.461	4.484
-79.196	-49.162
-7.688	18.219
15.823	45.630
-14.177	15.630
54.769	53.579
65.019	64.792
2.957	31.207

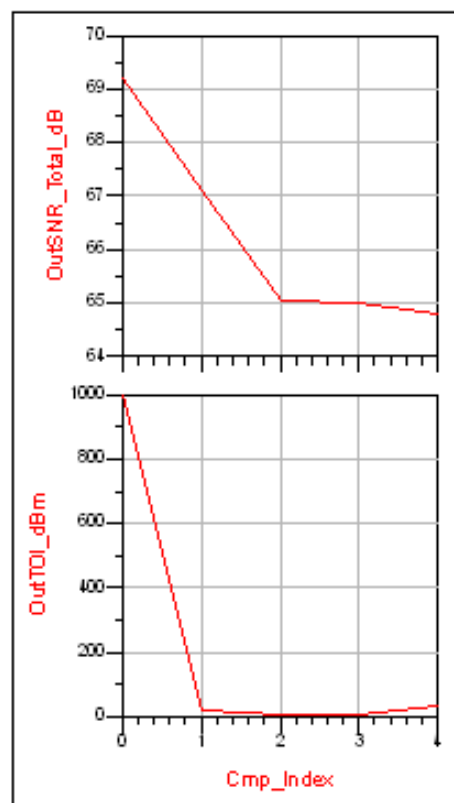
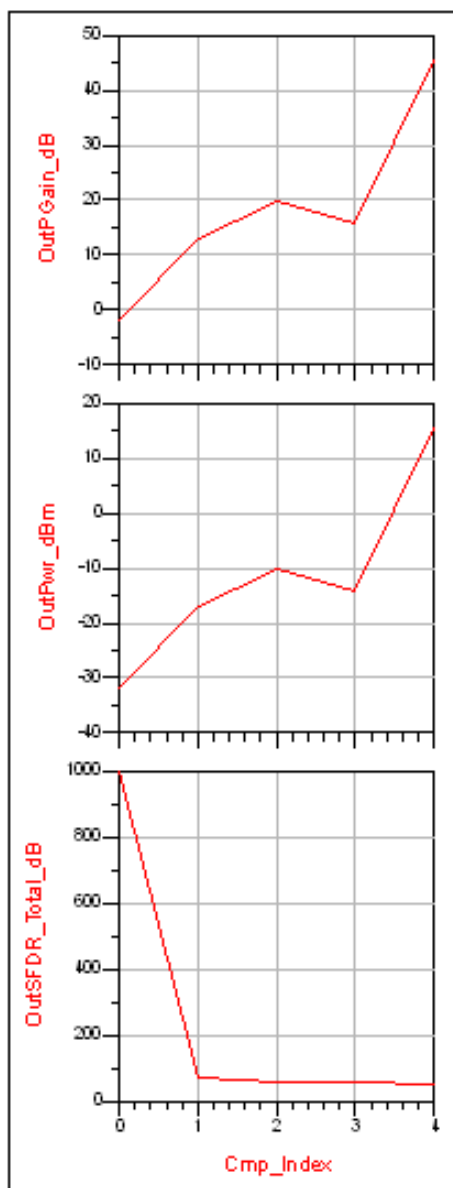
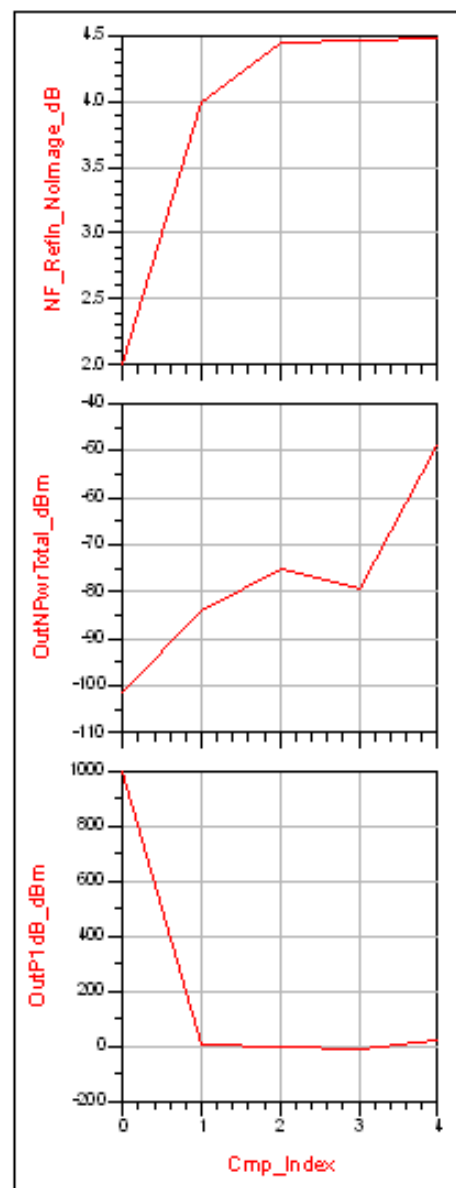
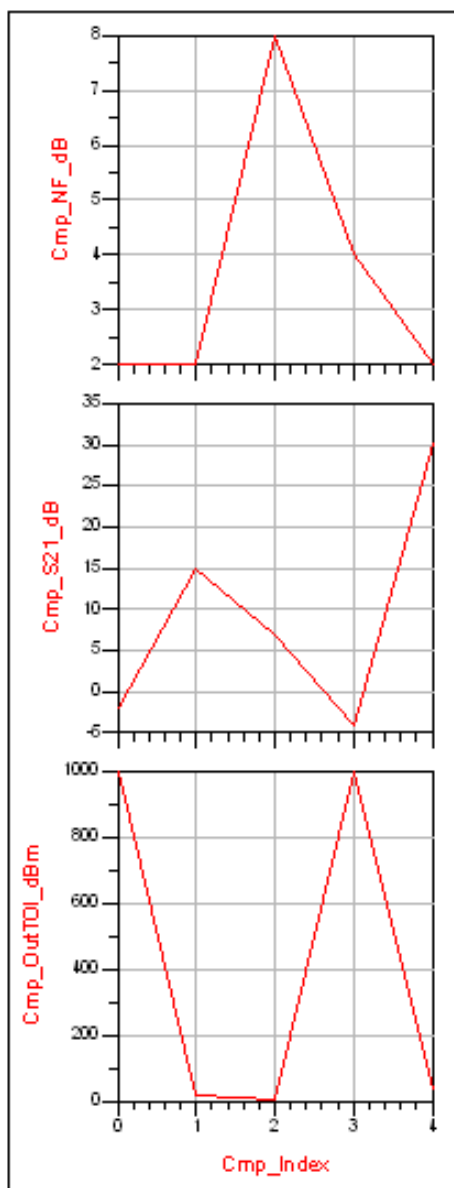
[Measurement tables](#)[Summary tables](#)[Measurement plots](#)



System_Name	System_Value
SystemInNO_dBm	-173.975
SystemInNPwr_dBm	-100.965
SystemInP1dB_dBm	-26.781
SystemInSOI_dBm	1000.000
SystemInTOI_dBm	-14.793
SystemNF_dB	6.054
SystemOutNO_dBm	-121.921
SystemOutNPwr_dBm	-49.162
SystemOutP1dB_dBm	18.219
SystemOutSOI_dBm	1000.000
SystemOutTOI_dBm	31.207
SystemPGain_SS_dB	46.000
SystemPGain_dB	45.630
SystemPOut_dBm	15.630
SystemS11_dB	-260.227
SystemS11_mag	9.742E-14
SystemS11_phase	89.527
SystemS12_dB	-400.000
SystemS12_mag	0.000
SystemS12_phase	0.000
SystemS21_dB	45.630
SystemS21_mag	191.201
SystemS21_phase	-7.804
SystemS22_dB	-313.360
SystemS22_mag	2.148E-16
SystemS22_phase	180.000

Setup_Name	Setup_Value
System_AnalysisType	1.000
System_NoiseResBW	1.000E6
System_NoiseSimBW	2.000E7
System_NoiseSimFStep	5.000E5
System_PilotFreq	2.000E9
System_PilotPwr_dBm	-30.000
System_RefR	50.000
System_SourceFreq	2.000E9
System_SourcePwr_dBm	-30.000
System_SourceTemp	16.850







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谢谢！

